ModelIt!

Teacher's Guide

What Happens When Habitats Change?

How Animals Survive Environmental Change

Quick Reference

Grade Level	Grade 3
NGSS Standard	3-LS4-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: 3-LS4-4

Make a claim about the merit of a solution to a problem caused when the environment changes and the types of plants and animals that live there may change.

Three-Dimensional Learning

Science Practice	**Engaging in Argument from Evidence** — Students make claims about the merit of solutions to problems caused by environmental changes, supporting their arguments with evidence from their model about habitat change effects.
Disciplinary Core Idea	**LS4.D Biodiversity and Humans** — Populations live in a variety of habitats, and change in those habitats affects the organisms living there. When the environment changes, some organisms survive, some move to new locations, and some die.
Crosscutting Concept	**Cause and Effect** — Students investigate how environmental changes (cause) lead to decreased food availability and lower animal populations (effects), then evaluate solutions to mitigate these effects.

CAST Testing Alignment

MULTIPLE CHOICE:

A drought dries up most of the water in a wetland. Which animal is MOST likely to survive — a frog that only lives in water, or a deer that can walk to find new water sources?

CONSTRUCTED RESPONSE:

A forest fire destroys half of a national park. Using your model, explain what happens to the animals that lived there and suggest one solution to help them survive. Use the words 'habitat change' and 'food availability' in your answer.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Habitats are the homes of every plant and animal on Earth. When these homes change — from natural events or human activity — the living things that depend on them face serious challenges to their survival.

What Is a Habitat?

A habitat provides four essential things every organism needs: food, water, shelter, and space. A coral reef provides all four for thousands of fish species. A forest provides them for birds, mammals, insects, and plants. Each species is adapted to its specific habitat — a polar bear is perfectly suited for Arctic ice but would struggle in a desert. When any of the four essentials is reduced or removed, the organisms in that habitat face survival challenges.

How Do Habitats Change?

Habitats change through both natural events and human activities. Natural changes include wildfires, droughts, floods, volcanic eruptions, and storms. Human-caused changes include deforestation, urban development, pollution, and climate change. Some changes are sudden (a fire can destroy a forest in days) while others are gradual (pollution can slowly poison a river over years). The speed and size of the change determines how much impact it has on the living things there.

How Do Animals Respond?

When habitats change, animals have three basic options: adapt, move, or perish. Some animals adapt by changing their behavior — eating different foods, using different shelters, or being active at different times. Some animals migrate to new habitats that still have what they need. Unfortunately, animals that cannot adapt or move may die. Generalist species (those that eat many foods and live in many habitats) survive changes better than specialist species (those that depend on one specific food or habitat).

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify habitat change as an external component and food availability and animal population as internal components that respond to how much the environment has changed.

E **Establish Relationships**
Students determine that habitat change negatively affects food availability, and food availability positively affects animal population — more habitat destruction means less food means fewer surviving animals.

V **Visualize & Model**
Students build a model showing how increasing habitat change leads to decreased food availability and lower animal populations.

E **Evaluate Outcomes**
Students run small change, moderate change, and major change scenarios to observe how the severity of habitat change affects food and population across the system.

R **Revise & Extend**
Students add shelter availability or species flexibility to explore why some animals survive habitat changes better than others.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Imagine you come home from school and your house has been completely rearranged — your bedroom is gone, the kitchen is different. How would you feel? That is what animals experience when habitats change!"*

Watch For

Show before-and-after photos of a real habitat change (forest fire, deforestation). Ask students to describe what changed.

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we are going to discover what happens to animals when their home — their habitat — suddenly changes!"*

Watch For

Read objectives together. Introduce vocabulary — show pictures of different habitats and name the populations living in each.

Slide 3: The Big Question

3 min

Say: *"When a forest burns or a river dries up, what happens to all the animals that lived there? Do they all make it?"*

Watch For

Turn and talk: What would you do if your home suddenly changed? Would all your neighbors survive the change?

Slide 4: Let's Build a Model!

3 min

Say: *"We are going to build a model to test what happens when habitats change — and find out which animals survive!"*

Watch For

Preview each LEVER step. Explain that conservation biologists study this exact question every day.

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"What parts of this system can we identify? What causes the changes, and what are the results?"*

Watch For

Guide sorting of three components. Explain habitat change is external because events like fires and droughts happen to the habitat from outside.

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When a habitat changes a LOT, does food go up or down? And what happens to animal population?"*

Watch For

Help students draw a negative arrow from habitat change to food availability, and a positive arrow from food to population.

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Let us test our model! What happens with a small change versus a disaster? Let us find out!"*

Watch For

Have students predict first, then run each scenario. Discuss which animals survived in each and why.

Slide 8: What Did We Discover?

7 min

Say: *"Habitat change is a real challenge for animals — but there ARE solutions we can use to help!"*

Watch For

Summarize discoveries. Discuss real-world solutions: wildlife corridors, replanting programs, protected areas. Ask: What can WE do?

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Habitat Change
- **INTERNAL:** Food Availability, Animal Population

Relationships

- **Habitat Change to Food Availability:** POSITIVE (+) — Greater habitat change destroys more of the food sources that animals depend on. A small change might remove a few food plants, but a major change like a wildfire can destroy most of the food supply. Note: this is actually a negative relationship in real terms — more change means LESS food — but the model shows that as habitat change increases, the impact on food availability increases.
- **Food Availability to Animal Population:** POSITIVE (+) — When more food is available, more animals can survive and the population stays healthy. When food decreases, animals compete, some starve, and the population drops. Food is the most important factor determining how many animals a habitat can support.

Key Discoveries

- When habitats change, food and shelter decrease, which causes animal populations to drop
- Some animals can adapt to changes — they move to new areas, change what they eat, or use different shelters
- Animals that are specialized (eat only one food or live in only one type of place) are most at risk when habitats change
- Humans can help by creating wildlife corridors, replanting forests, and protecting remaining habitats

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Animals can just move somewhere else when their habitat changes"	Moving is not easy for all animals. Some animals (like turtles and salamanders) move very slowly. Some (like specialized insects) need very specific plants that only grow in certain places. And even fast animals face dangers when crossing roads, farmland, or cities to reach new habitat. Moving is not always an option.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Nature always bounces back quickly after changes"	Some habitats recover quickly (grasslands after fire can regrow in one year), but others take decades or centuries. Old-growth forests that took 500 years to develop cannot be replaced quickly. Coral reefs damaged by pollution may take 25-75 years to recover. Some ecosystems, once destroyed, may never fully return.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Only big changes matter — small changes do not affect animals"	Small, repeated changes can add up to major impacts over time. Removing a few trees each year seems small, but after 20 years, half the forest is gone. Pollution that is barely noticeable each day can build up to toxic levels over months. Scientists call this 'death by a thousand cuts' — many small changes can be just as devastating as one big change.